

Cover letter for manuscript
“Efficient CP Rounding using Alternating Least Squares with
QR Decomposition”

Dear Editors,

With this letter, we are submitting a manuscript entitled “Efficient CP Rounding using Alternating Least Squares with QR Decomposition” for consideration in the SIAM Journal on Matrix Analysis and Applications.

The main contributions of this manuscript are summarized in the abstract, which we paraphrase here for your convenience:

We present an efficient QR-based alternating least squares (ALS) algorithm for CP rounding. CP rounding is the problem of computing a lower-rank CP decomposition of an input already given in a higher-rank CP format. The normal equations (NE) method in CP-ALS is commonly used and efficient for solving the CP rounding problem; however, it becomes unstable in the presence of ill-conditioned subproblems. Our QR-based approach addresses this issue and provides a numerically stable algorithm that has computational complexity similar to the NE-based method. Our experiments show that the proposed method offers significant speedup compared to a previous QR-based approach and the NE-based implementation in the Tensor Toolbox, especially for higher-order tensors. Moreover, our approach leads to substantial error reductions in ill-conditioned problems, with error reductions several orders of magnitude smaller compared to the NE-based method, while achieving faster convergence and more accurate solutions.

We believe that the novel results in this manuscript will be of interest to a significant portion of your readership.

We hope that you will strongly consider this submission for publication, and we eagerly await your response.

Best regards,
Alex Zhang, Bhisham Dev Verma, Jan Van Lent, and Grey Ballard